Improving Hallucination Detection in Dialogue

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MOTIVATION

- **Context-Driven Failures:** Models invent speaker names, fabricate details, and fill unstated gaps with assumptions.
- **Multi-Turn Degradation:** Longer conversations cause models to lose earlier turns and get distracted by irrelevant context, lowering detection accuracy.
- **Lack of Dialogue-Level Data:** Most benchmarks are sentence- or passage-level; dialogue benchmarks have outdated annotations or cover only last-turn factual hallucinations.

Factuality

How tall is Mount Everest?

✗ About 5,000 feet.

Shorter than I thought!

Yes, it surprises many people!

Faithfulness

I'm a vegan, I believe in treating all creatures with kindness.

That's great! I'm vegan too.

Shall we go get some beef tacos?

✗ That sounds delicious! I'm always up for good food.

DIALOGUE TRANSFORMATIONS

- **Dataset:** DiaHalu, - **Transformer LLM**: GPT4o mini, - **Detector LLMs**: Claude-Haiku-4.5, GPT-5.4-nano, GPT-5.4-mini, Qwen3-8B, Gemma-3-4B, Gemma-3-12B, Gemma-3-27B

Base dialogue:

A: Have you heard this slogan? I can't remember what drinking Red Bull gives you.
B: It can give you wings!

Surface Transforms

S1 Coreference resolution:

A: Have you heard this slogan? I can't remember what drinking Red Bull gives you. **B:** **Drinking Red Bull** can give you wings!

Win: 0, Lose: 0, NS: 7

S2 Formalization:

A: Have you heard this slogan? I **cannot recall** what drinking Red Bull **provides**.
B: It can **provide you with** wings.

Win: 0, Lose: 3, NS: 4

S3 Turn concatenation:

Have you heard this slogan? I can't remember what drinking Red Bull gives you. It can give you wings!

Win: 2, Lose: 0, NS: 5

S4 Dehumanization:

AI Agent 1: Have you heard this slogan? I can't remember what drinking Red Bull gives you. **AI Agent 2:** It can give you wings!

Win: 2, Lose: 0, NS: 5

Content Transforms

F1 Flattening to prose:

Person A cannot remember what drinking Red Bull gives you. In regard to the slogan, it can give you wings.

Win: 1, Lose: 2, NS: 4

F2 Atomic claim decomposition:

There is a slogan associated with drinking Red Bull.
The speaker cannot remember what drinking Red Bull gives you.
Drinking Red Bull can give you wings.

Win: 1, Lose: 3, NS: 3

F3 Dialogue summarization:

A1 mentioned not being able to remember the slogan related to drinking Red Bull. B1 responded that drinking Red Bull can give you wings.

Win: 0, Lose: 2, NS: 5

F4 QA conversion:

Q: What does drinking Red Bull give you? **A:** Drinking Red Bull can give you wings.

Win: 1, Lose: 4, NS: 2

Dialog Native Approaches

Zero Shot (ZS)

Full dialogue, 1 call.

A1: *user turn 1*
B1: *response turn 1*
A2: *user turn 2*
B2: *response turn 2*

Does the dialogue contain a hallucination?

Inline Checkpoints (Ckp)

>> checkpoints after each full exchange, 1 call.

A1: *user turn 1*
B1: *response turn 1*

Does the dialogue contain a hallucination?

A2: *user turn 2*
B2: *response turn 2*

Does the dialogue contain a hallucination?

Checkpoints + Repetition (Ckp+Rep)

>> Checkpoints with history repeated at each step, 1 call.

A1: *user turn 1*
B1: *response turn 1*

Does the dialogue contain a hallucination?

HISTORY:
A1: *user turn 1*
B1: *response turn 1*

NEW TURNS:
A2: *user turn 2*
B2: *response turn 2*

Does the dialogue contain a hallucination?

Progressive Detection (Exch)

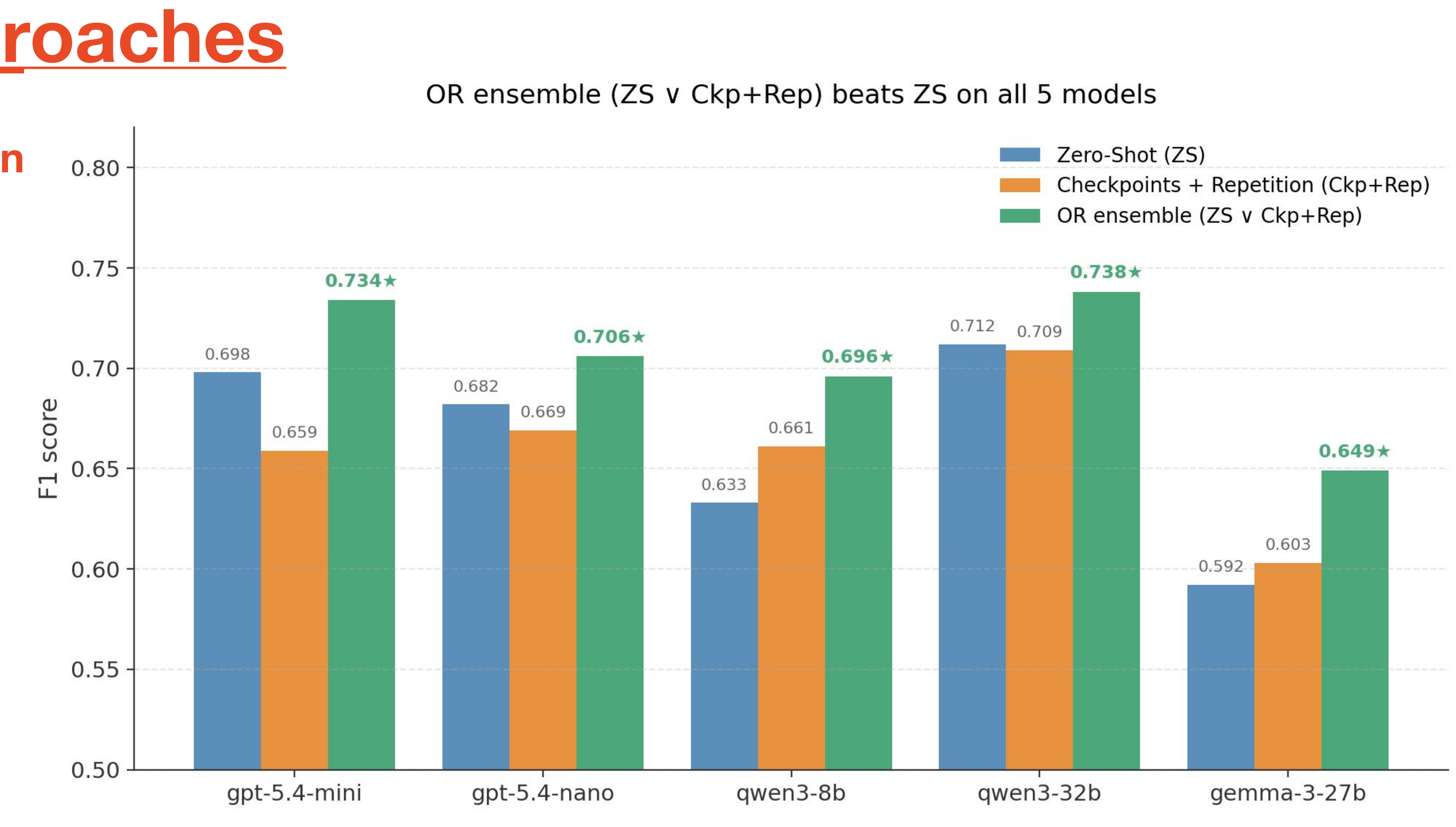
N independent calls, one per prefix; OR-aggregate.

A1: *user turn 1*
B1: *response turn 1*

Does the dialogue contain a hallucination?

A1: *user turn 1*
B1: *response turn 1*
A2: *user turn 2*
B2: *response turn 2*

Does the dialogue contain a hallucination?

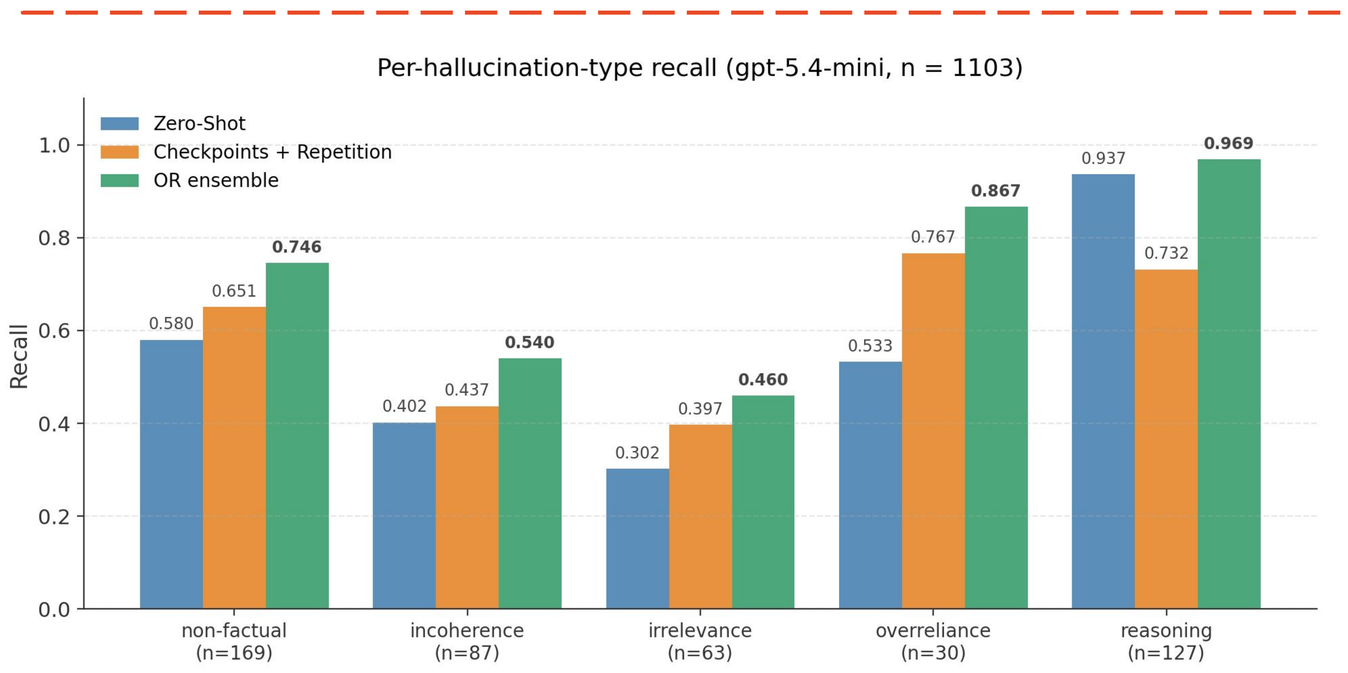


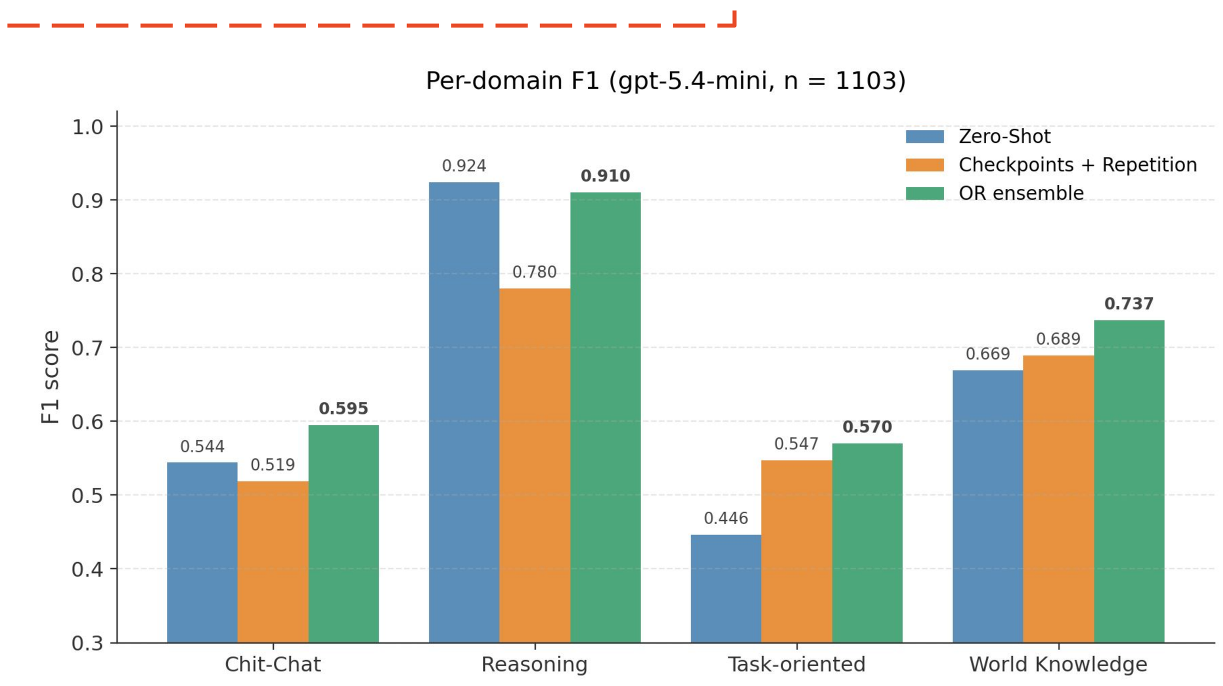
Findings:

- Inline Checkpoints (Ckp), Progressive Detection (Exch), and Ckp+Rep **alone** do not beat ZS on their own (subset, n=199).
- Ckp+Rep alone is the weakest individual detector, yet makes the strongest ensemble partner — it flags errors ZS misses.
- 64 hallucinations caught only by Ckp+Rep; 62 only by ZS. Near-symmetric disagreement confirms genuine complementarity.

Finding:

OR-ensembling Zero-Shot with Checkpoints+Repetition yields significant F1 gains on all 5 models (+3 to +6 F1), while neither approach wins alone.





Finding:

ZS and Ckp+Rep are **complementary**, not redundant. ZS dominates reasoning errors (+21 recall), Ckp+Rep wins on dialogue-specific errors like overreliance (+23 recall) and irrelevance (+10 recall); the ensemble recovers both.

TAKEAWAYS

- **Dialogue structure matters.** Transforms that simplify or remove it (F1–F4, S2) hurt detection; transforms that preserve it (S3, S4, S7) help.
- **Single-pass detection has blind spots.** ZS misses dialogue-specific errors (overreliance, irrelevance) that exchange-level auditing catches.
- **Complementarity beats sophistication.** The strongest detector is not a new method, but the OR ensemble of the simplest (ZS) and a dialogue-native auditor (Ckp+Rep).

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